

ROW-INCLUSION RESIDUAL DYNAMICS ON BOOLEAN RELATIONS: PREORDER TWO-CYCLES AND EXACT FIBRES

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ABSTRACT. For an $n \times n$ Boolean matrix A , replace A by the matrix that records inclusion among its row supports. The first image is always a labelled preorder, and on every preorder the update is transpose. We derive the global identity $\mathsf{T}_n^3 = \mathsf{T}_n$: nonpreorders have tail one, equivalence relations are exactly the fixed points, and all other preorders form strict transpose two-cycles. This gives every temporal fixed-set count and the dynamical zeta function. We also solve the one-step inverse problem for every target. For a preorder P , preimages are precisely induced order embeddings of its antisymmetric quotient into the Boolean lattice $\mathcal{B}_n$. Encoding Boolean coordinates as upper sets yields an explicit inclusion–exclusion formula for $|\mathsf{T}_n^{-1}(P)|$; nonpreorder targets have empty fibre. Relational self-residuation and its preorder property are standard background and receive no contribution credit. Exact exhaustion through $n = 4$ is used only for falsification, and external release remains on hold.

1. THE LITERAL MAP AND THEOREM

Let $\mathcal{B}_n = 2^{[n]}$ be the Boolean lattice. For a Boolean matrix $A \in \{0, 1\}^{n \times n}$, write

$$R_i(A) = \{j \in [n] : A_{ij} = 1\}.$$

Define the literal self-map

$$(1) \quad \mathsf{T}_n(A)_{ij} = 1 \iff R_i(A) \subseteq R_j(A).$$

We represent a preorder P on $[n]$ by the Boolean matrix with $P_{ij} = 1$ exactly when $i \leq_P j$. Let q_n denote the number of labelled preorders on $[n]$, and let B_n denote the n th Bell number.

For a finite preorder S , let

$$J(S) = |\mathsf{Up}(S)|$$

be its number of upper sets. Given a preorder P , define $i \sim_P j$ when $i \leq_P j$ and $j \leq_P i$, and let $Q = P/\sim_P$ be the resulting quotient poset. Put

$$(2) \quad D_Q = \{(q, r) \in Q^2 : q \not\leq_Q r\}.$$

For $S \subseteq D_Q$, let Q_S be the preorder obtained by adjoining every pair in S to Q and taking reflexive–transitive closure.

Theorem 1.1 (Complete forward and inverse dynamics). *For every $n \geq 1$, the following hold.*

(i) *The image of T_n is precisely the set of labelled preorders, and*

$$(3) \quad \mathsf{T}_n(P) = P^\top$$

for every preorder P . In particular,

$$(4) \quad \mathsf{T}_n^3 = \mathsf{T}_n.$$

(ii) *Every nonpreorder has tail exactly one. The fixed points are exactly the equivalence relations; every other preorder belongs to a strict transpose two-cycle. Consequently, for every $k \geq 1$,*

$$(5) \quad |\mathsf{Fix}(\mathsf{T}_n^k)| = \begin{cases} B_n, & k \text{ odd}, \\ q_n, & k \text{ even}, \end{cases}$$

and the Artin–Mazur zeta function is

$$(6) \quad \zeta_{\mathsf{T}_n}(z) = (1 - z)^{-B_n} (1 - z^2)^{-(q_n - B_n)/2}.$$

(iii) *A target outside the preorders has empty fibre. For every preorder P ,*

$$(7) \quad |\mathsf{T}_n^{-1}(P)| = \sum_{S \subseteq D_Q} (-1)^{|S|} J(Q_S)^n.$$

Equivalently, the fibre is the set of induced order embeddings $Q \hookrightarrow \mathcal{B}_n$.

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The operation in (1) is a row-oriented instance of relational self-residuation. Relation algebra and containment preorders are standard; see Schmidt's monograph [3]. Likewise, induced embeddings in Boolean lattices belong to an established order-theoretic literature [2]. Our scope is the iterated finite map and its complete inverse atlas, not invention of either construction.

2. COLLAPSE TO A PREORDER AND ORDER REVERSAL

Lemma 2.1 (The residual preorder). *For every Boolean matrix A , the relation $\mathsf{T}_n(A)$ is reflexive and transitive.*

Proof. Every row support contains itself, so $R_i(A) \subseteq R_i(A)$ gives reflexivity. If $R_i(A) \subseteq R_j(A)$ and $R_j(A) \subseteq R_k(A)$, then $R_i(A) \subseteq R_k(A)$, giving transitivity. $\square$

Lemma 2.2 (Principal upper sets reverse inclusion). *If P is a preorder, then $\mathsf{T}_n(P) = P^\top$.*

Proof. Row i of P is the principal upper set $\uparrow i = \{x : i \leq_P x\}$. We claim

$$\uparrow i \subseteq \uparrow j \iff j \leq_P i.$$

If the inclusion holds, reflexivity puts i in $\uparrow i$, hence in $\uparrow j$, so $j \leq_P i$. Conversely, $j \leq_P i \leq_P x$ implies $j \leq_P x$ by transitivity. Substitution in (1) proves the lemma. $\square$

Proof of theorem 1.1(i)–(ii). By lemma 2.1, the image is contained in the preorders. Conversely, if P is a preorder then so is $P^\top$, and lemma 2.2 gives

$$\mathsf{T}_n(P^\top) = P.$$

Thus every preorder occurs in the image. Applying lemma 2.2 twice after the first update gives (4).

A state is periodic only if it lies in the image, hence only if it is a preorder. On the preorder image the map is transpose, so every period is one or two. A preorder is fixed exactly when it is symmetric. A reflexive, transitive, symmetric relation is an equivalence relation, and the converse is immediate. Equivalence relations on $[n]$ are counted by B_n ; the remaining $q_n - B_n$ preorder states pair into strict cycles.

Equation (5) follows directly. Finally, using $\zeta_{\mathsf{T}_n}(z) = \exp(\sum_{k \geq 1} |\text{Fix}(\mathsf{T}_n^k)| z^k / k)$, or multiplying one factor for each cycle, yields (6). $\square$

The numbers q_n also count finite labelled topologies, a classical correspondence and enumeration problem [1]. We use this census only to state the periodic population; it is not part of the residual claim.

3. QUOTIENT-POSET GEOMETRY OF A FIBRE

Proposition 3.1 (Fibre as an induced embedding set). *Let P be a preorder and $Q = P / \sim_P$. Sending a source matrix A to its row-support map induces a bijection*

$$(8) \quad \mathsf{T}_n^{-1}(P) \longleftrightarrow \{f : Q \hookrightarrow \mathcal{B}_n : q \leq_Q r \iff f(q) \subseteq f(r)\}.$$

Proof. Suppose $\mathsf{T}_n(A) = P$. By definition,

$$(9) \quad i \leq_P j \iff R_i(A) \subseteq R_j(A).$$

If $i \sim_P j$, both inclusions in (9) hold and the rows are equal. Conversely, equal rows force both relations, so the row assignment descends to an injective map $f : Q \rightarrow \mathcal{B}_n$. The two directions in (9) say exactly that f preserves and reflects order.

Conversely, from an induced order embedding f assign row support $f([i])$ to every i . Order preservation and reflection then make (9) hold, so the resulting Boolean matrix maps to P . The two constructions are inverse. $\square$

The quotient is essential. Mutual comparability in a preorder is exactly the equality forced among source rows; treating all n labels as distinct would incorrectly demand an injective embedding before this forced collapse.

4. EVERY-TARGET INCLUSION-EXCLUSION

Lemma 4.1 (Coordinate encoding). *For any finite preorder S , the number of order-preserving maps $f : S \rightarrow \mathcal{B}_n$ is $J(S)^n$.*

Proof. For coordinate $c \in [n]$, put

$$U_c = \{q \in S : c \in f(q)\}.$$

The map f is order preserving exactly when every U_c is an upper set. The n coordinates are independently chosen, giving $J(S)^n$. Conversely, the chosen upper sets recover $f(q) = \{c : q \in U_c\}$. $\square$

Proof of theorem 1.1(iii). A nonpreorder cannot be a target by lemma 2.1. Fix a preorder P and its quotient poset Q . Begin with all order-preserving maps $f : Q \rightarrow \mathcal{B}_n$. For each $(q, r) \in D_Q$, let $E_{q,r}$ be the bad event

$$f(q) \subseteq f(r).$$

Avoiding every such event is precisely order reflection, and hence, by proposition 3.1, membership in the desired fibre.

If all events indexed by $S \subseteq D_Q$ are imposed, then f must preserve the original order of Q together with every adjoined comparison in S . This is equivalent to being order preserving for the reflexive–transitive closure Q_S . By lemma 4.1, the number of such maps is $J(Q_S)^n$. Inclusion–exclusion over the bad events gives (7). $\square$

The formula remains valid when added comparisons identify elements in Q_S : upper sets depend only on the resulting preorder. Standard inclusion–exclusion and poset enumeration are used in their ordinary senses [4]; no algorithmic novelty is asserted.

5. EXACT AUDIT AND LIMITATIONS

The paper-local verifier exhausts all 2^{n^2} matrices for $n \leq 4$. For each source it independently checks preorder formation, $\mathsf{T}(P) = P^\mathsf{T}$, and $\mathsf{T}^3 = \mathsf{T}$. It then compares the formula (7) with direct fibre counts for every preorder target. Table 1 is a compact fingerprint of the frozen run.

TABLE 1. Exact theorem–interface audit. “Strict states” counts states in strict two-cycles, not cycles.

n	states	preorders	fixed	strict states	min fibre	max fibre
1	2	1	1	0	2	2
2	16	4	2	2	2	5
3	512	29	5	24	8	24
4	65,536	355	15	340	16	600

The run contains 265,050 exact assertions. Enumeration supplies no step of the proofs and says nothing about novelty beyond the tested boxes.

There are also substantive limitations. Formula (7) is exact but exponential in the number of missing ordered pairs; we do not claim an efficient general evaluator. It does not simplify the still-difficult global enumeration of labelled preorders. Most importantly, the first-step self-residual construction is owned relation-algebra background. A bounded source search did not locate the precise iterated-plus-inverse conjunction, but a non-hit is not a priority or novelty certificate.

6. CONCLUSION

Row-support inclusion sends the full Boolean-relation cube onto the preorder stratum in one step, and principal upper sets turn the next update into order reversal. This produces a complete one-step-tail, one/two-cycle dynamical atlas. Quotienting mutual comparability exposes the inverse geometry: source rows are induced Boolean-lattice representations, and upper-set coordinates make every fibre accessible to inclusion–exclusion. These statements form an internally verified theorem package; external release, priority, authorship, and submission remain on hold.

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